



SIMATS Online Programming Lab

DS PROGRAMMING



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QUESTIONS

10 / 10 completed

Single Sorted Linked List

SINGLE SORTED LINKED LIST

"Two departments maintain separate sorted lists of student IDs. For centralized processing, both lists need to be merged while preserving the sorted order. Write a C program to merge two sorted singly linked lists into a single sorted linked list.

Test Case 1

Input

List 1: 10 → 30 → 50

List 2: 20 → 40 → 60

Output

10 → 20 → 30 → 40 → 50 → 60 → NULL"

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node* next;
7 };
8
9 struct Node* createNode(int data) {
10     struct Node* newNode = (struct Node*)malloc(sizeof(struct N
11     if (!newNode) {
12         printf("Memory allocation failed.\n");
13         exit(1);
14     }
15     newNode->data = data;
16     newNode->next = NULL;
17     return newNode;
18 }
19
20 void appendNode(struct Node** head, int data) {
21     struct Node* newNode = createNode(data):
```

OUTPUT

Merged List:

10 -> 20 -> 30 -> 40 -> 50 -> 60 -> NULL

INPUT

10 30 50

20 40 60

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